

# CSEGriskfigure

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CSEGriskfigure

*Plot Extinction Risk Metrics*

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## Description

Generates a six-panel plot of extinction risk metrics used in Population Viability Analysis (PVA). This is a function used by one of the vignettes in the [MARSS-package\(\)](#).

## Usage

```
CSEGriskfigure(  
  data,  
  te = 100,  
  absolutethresh = FALSE,  
  threshold = 0.1,  
  datalogged = FALSE,  
  silent = FALSE,  
  return.model = FALSE,  
  CI.method = "hessian",  
  CI.sim = 1000  
)
```

## Arguments

|                |                                                                                                                                                                                     |
|----------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| data           | A data matrix with 2 columns; time in first column and counts in second column. Note time is down rows, which is different than the base <a href="#">MARSS-package()</a> functions. |
| te             | Length of forecast period (positive integer)                                                                                                                                        |
| absolutethresh | Is extinction threshold an absolute number? (T/F)                                                                                                                                   |
| threshold      | Extinction threshold either as an absolute number, if absolutethresh=TRUE, or as a fraction of current population count, if absolutethresh=FALSE.                                   |
| datalogged     | Are the data already logged? (T/F)                                                                                                                                                  |
| silent         | Suppress printed output? (T/F)                                                                                                                                                      |

|                           |                                                                                                                   |
|---------------------------|-------------------------------------------------------------------------------------------------------------------|
| <code>return.model</code> | Return state-space model as <code>marssMLE()</code> object? (T/F)                                                 |
| <code>CI.method</code>    | Confidence interval method: "hessian", "parametric", "innovations", or "none". See <code>MARSSparamCIs()</code> . |
| <code>CI.sim</code>       | Number of simulations for bootstrap confidence intervals (positive integer).                                      |

### Details

- Panel 1: Time-series plot of the data.
- Panel 2: CDF of extinction risk.
- Panel 3: PDF of time to reach threshold.
- Panel 4: Probability of reaching different thresholds during forecast period.
- Panel 5: Sample projections.
- Panel 6: TMU plot (uncertainty as a function of the forecast).

### Value

If `return.model=TRUE`, an object of class `marssMLE()`.

### Author(s)

Eli Holmes, NOAA, Seattle, USA, and Steve Ellner, Cornell Univ.

### References

Holmes, E. E., E. J. Ward, and M. D. Scheuerell (2012) Analysis of multivariate time-series using the MARSS package. NOAA Fisheries, Northwest Fisheries Science Center, 2725 Montlake Blvd E., Seattle, WA 98112 Type `RShowDoc("UserGuide", package="MARSS")` to open a copy.

(theory behind the figure) Holmes, E. E., J. L. Sabo, S. V. Viscido, and W. F. Fagan. (2007) A statistical approach to quasi-extinction forecasting. *Ecology Letters* 10:1182-1198.

(CDF and PDF calculations) Dennis, B., P. L. Munholland, and J. M. Scott. (1991) Estimation of growth and extinction parameters for endangered species. *Ecological Monographs* 61:115-143.

(TMU figure) Ellner, S. P. and E. E. Holmes. (2008) Resolving the debate on when extinction risk is predictable. *Ecology Letters* 11:E1-E5.

### See Also

`MARSSboot()`, `marssMLE()`, `CSEGriskfigure()`

### Examples

```
d <- harborSeal[, 1:2]
kem <- CSEGriskfigure(d, datalogged = TRUE)
```

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